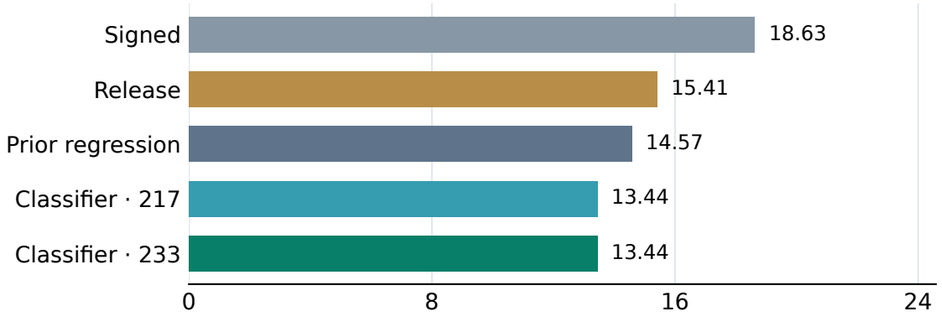
Learning hardware-budget feasibility directly

Two matched 300-tree classifiers · identical 1,889 training circuits · fixed feasibility threshold of 0.5

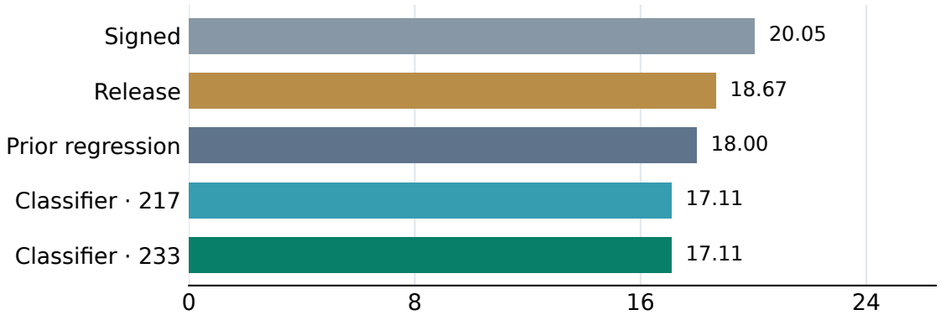
Development selection: Classifier · 233 · Evaluated: Classifier · 233 · Not eligible for promotion

General cohort · 32 fixed circuits

Policy MAE · LUT4 ↓



Policy positive-target MAPE · % ↓



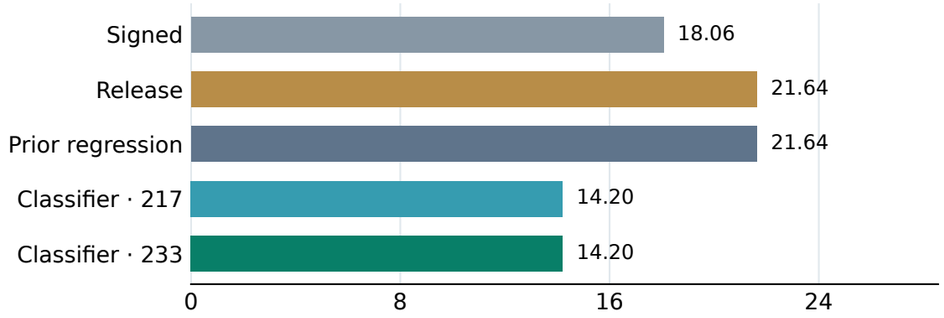
False accepts / false rejects · each budget assessed separately

Cap	Signed	Release	Prior	217	233
32	0 / 2	0 / 2	0 / 2	0 / 2	0 / 2
64	3 / 0	1 / 0	1 / 0	1 / 0	1 / 0
128	6 / 0	1 / 5	1 / 2	1 / 0	1 / 0

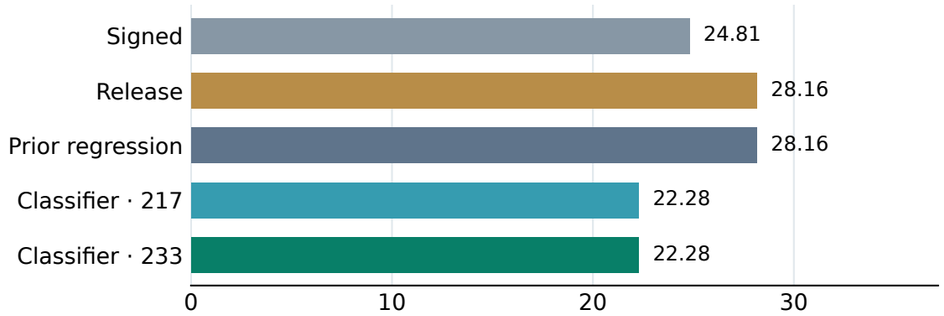
Feasible / infeasible: 32: 2/30 · 64: 7/25 · 128: 20/12

Conditional cohort · 8 active + 8 shared controls

Policy MAE · LUT4 ↓



Policy positive-target MAPE · % ↓



False accepts / false rejects · each budget assessed separately

Cap	Signed	Release	Prior	217	233
32	0 / 2	0 / 2	0 / 2	0 / 2	0 / 2
64	1 / 0	0 / 0	0 / 0	0 / 0	0 / 0
128	6 / 0	1 / 5	1 / 5	4 / 0	4 / 0

Feasible / infeasible: 32: 2/14 · 64: 2/14 · 128: 10/6

Mandatory sharing cases · n=8 · release → evaluated candidate: MAE 22.30 → 22.30 LUT4 · FA/FR at 128: 1/1 → 1/1

Mandatory legacy-disagreement cases · n=8 · release → evaluated candidate: MAE 25.39 → 25.16 LUT4 · FA/FR at 64: 1/0 → 1/0

Failed validation: 3 new active-case false accepts at 128; 1 sharing-case false rejection at 128.

Fresh activity: 8 action(s) / 2 template(s); legacy-64 vetoes: 1; sharing-128 vetoes: 1. Eight controls shared; no pooling.

Measured target: isolated iCE40 LUT4 count. This is not a CPU speedup, application-transfer or universal hardware-safety result.